

A blue trapezoidal graphic element on the left side of the title.

FA report of Greenish display (NDF)

JUNGLIM G133HAN01.1

Problem Statement



- General Information

Brand/Project	JUNGLIM				Defect Phenomenon	Greenish display(NDF)	
ES	YH				Defect Rate or Q'ty	2pcs	
AUO Model	G133HAN01.1				Defect Station	LINE REJECT	
AUO S/N	305B9B07RKZZZS0101. 305B8B944PZZZS0101				RMA No	702610101722	
Beol/Module	ARRAY_FAB_ID	FEOL_FAB_ID	CF_FAB_ID	BEOL_FAB_ID	LCM_BEOL_FAB_ID	JI_FAB_ID	MODULE_FAB_ID
	L5C	L5C	C5C	L5C	S01		S01
Week Code	25/44				Issue Date	2026.04.29	

- Defect Panel S/N Picture



Problem Statement

- Describe the phenomenon at customer side

The customer reported a greenish display phenomenon observed at the end system.

No abnormality was initially identified during standard outgoing inspection.

- Defect Phenomenon

Under normal display conditions (e.g., L92 pattern), no visible abnormality is observed. The panel appears uniform, and no greenish display is detected under this condition.



L92



no abnormality

L92

Root Cause Analysis

Step 1: Panel Verification

- The panel was evaluated under the **L64 pattern** and **gray-scale (gradient)** patterns.
- No visible abnormality or greenish phenomenon was observed during these tests.
- The display appearance remained uniform across the panel under both test conditions.



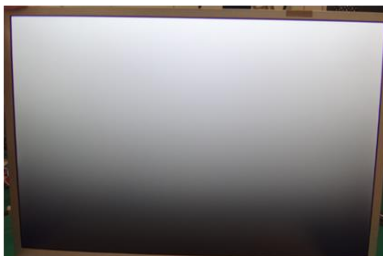
L64



L64



Gray-scale pattern



Gray-scale pattern

Remark:

The **L64 pattern** applied in this evaluation is a tightened inspection condition that was established based on a similar phenomenon observed previously, in agreement with the customer.

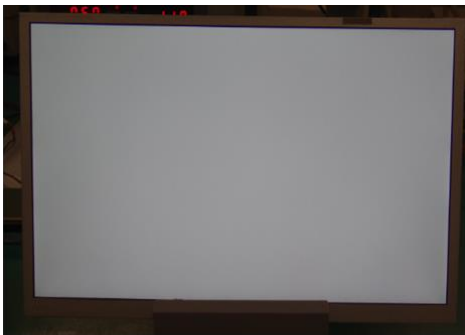
Root Cause Analysis

Step 2: Panel Verification (L128 Pattern)

- The panel was evaluated under the L128 pattern.
- No visible abnormality or greenish phenomenon was observed during this test.
- The display appearance remained uniform across the panel.



L128



L128



Root Cause Analysis

Step 3: Visual Inspection

- The panel appearance was visually inspected on both the front and rear side.
- No obvious abnormality, contamination, or defect was identified during the inspection.



Root Cause Analysis

Step 4: ITO Resistance Verification

- ITO sheet resistance (RS) measurements were performed on two sampling locations (#1 and #3) for each returned unit. Max., Min., and Avg. values were recorded at different probe distances.
- The measurement results show that the RS values are within a consistent range across different positions and distances, with no significant variation observed.



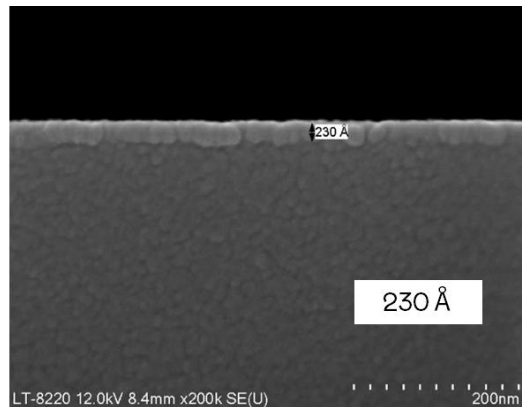
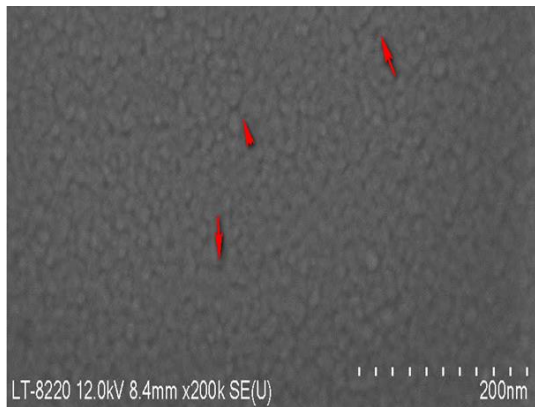
305B9B07RKZZZS010		量測位置	
探針距離	阻抗	#1	#3
0.5cm	Max	1.12	1.03
	Min	0.95	0.98
	Avg	1.03	1.01
1.0cm	Max	1.05	1.08
	Min	0.89	0.95
	Avg	0.98	1.01
1.5cm	Max	1.01	1.12
	Min	0.96	0.99
	Avg	0.99	1.05
2.0cm	Max	1.04	1.06
	Min	0.98	1.02
	Avg	1.01	1.04

305B8B944PZZZS0101		量測位置	
探針距離	阻抗	#1	#3
0.5cm	Max	0.97	1.02
	Min	0.91	0.94
	Avg	0.94	0.98
1.0cm	Max	1.03	1.06
	Min	0.98	1.01
	Avg	1.01	1.04
1.5cm	Max	1.02	1.07
	Min	0.96	0.89
	Avg	0.99	0.98
2.0cm	Max	1.05	1.08
	Min	0.96	0.97
	Avg	1.01	1.03

Root Cause Analysis

Step 5: BS-ITO SEM Analysis

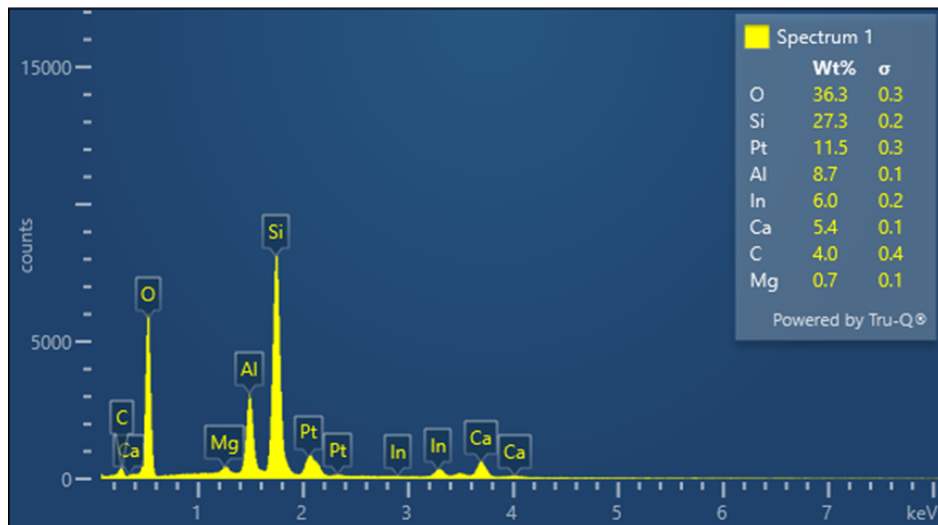
- SEM analysis was performed to examine the surface morphology and cross-sectional structure of the BS-ITO layer.
- The BS-ITO surface shows generally uniform morphology.
- Minor micro-scale features (as indicated in the images) were observed locally.
- The cross-sectional structure shows a thickness of approximately 230 Å.



Root Cause Analysis

Step 6: BS-ITO EDX Analysis

- EDX analysis was performed to examine the elemental composition of the BS-ITO layer.
- The results show that oxygen content is approximately 36.3 wt%.



oxygen content is approximately 36.3 wt%

Conclusion



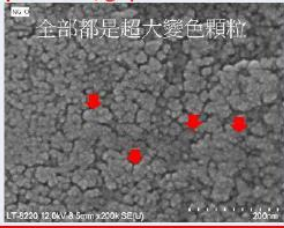
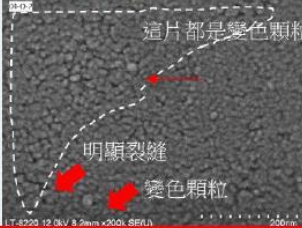
Conclusion and Countermeasure

1. No greenish phenomenon was observed under panel-level verification, including L64, L92, L128, and gray-scale patterns.
2. Visual inspection confirmed that no abnormality or defect was identified on both the display surface and module structure.
3. ITO resistance measurements show stable and consistent electrical performance, with no abnormal variation observed.
4. SEM and EDX analyses indicate that the BS-ITO structure and composition are within the expected range, with no abnormal characteristics identified.
5. Compared to the previous NG samples, the current samples exhibit a more controlled ITO thickness and lower oxygen content.

Item	Previous NG Samples (2025)	Current Samples (2026)
ITO Thickness	287–300 Å (out of expected range)	~230 Å (within expected range)
Oxygen Content (O wt%)	46.27% (elevated)	36.3% (controlled)
Sheet Resistance (RS)	Higher than expected	Within normal range
Observed Phenomenon	Clearly observed NG	Not reproducible

Appendix- I

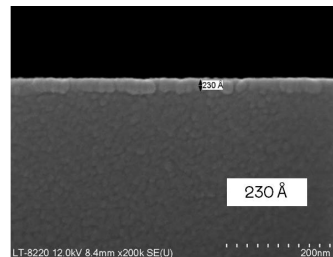
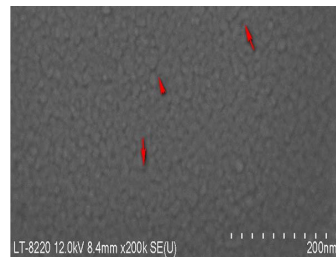
1. Compared to the previous evaluation results (supplier C) reported in 2025, the current samples from supplier A demonstrate comparable or more stable electrical performance , with a narrower RS distribution and no abnormal variation observed.
2. In addition, no significant structural abnormality is identified from SEM observation compared to the previous case.

	NG sample #4 (Supplier C Jan /no Polish)	NG sample #3 (Supplier C Jan /no Polish)	S01 inventory (Supplier C June /no Polish)
BSITO RS	1.5~1.8KΩ	1.2~1.3 KΩ	0.9~1.0KΩ
SEM	BSITO obvious particles/gaps 	BSITO obvious particles/gaps 	BSITO obvious particles/gaps 

2025 (supplier C)

105B9807RKZ2Z5010		樣品位置	
樣品距離	距離	#1	#3
0.5cm	Max	1.12	1.03
	Min	0.95	0.98
	Avg	1.05	1.01
1.0cm	Max	1.05	1.08
	Min	0.89	0.95
	Avg	0.98	1.01
1.5cm	Max	1.01	1.12
	Min	0.96	0.99
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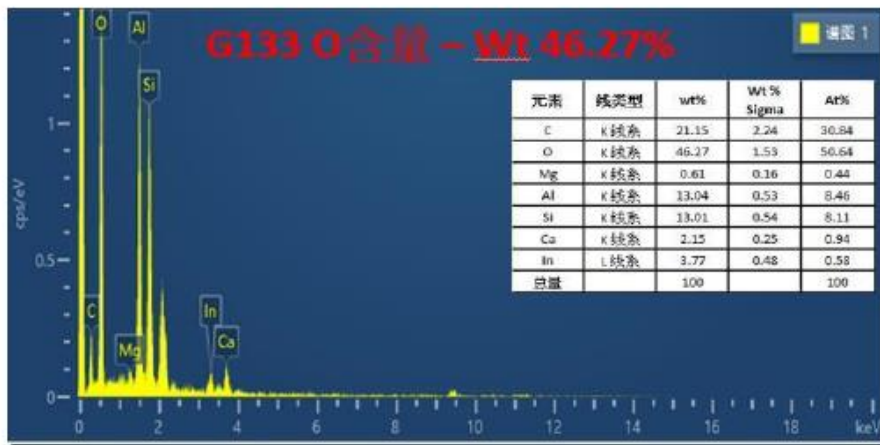
105B88944PZ2Z50101		樣品位置	
樣品距離	距離	#1	#3
0.5cm	Max	0.97	1.02
	Min	0.91	0.94
	Avg	0.94	0.98
1.0cm	Max	1.03	1.06
	Min	0.98	1.01
	Avg	1.01	1.04
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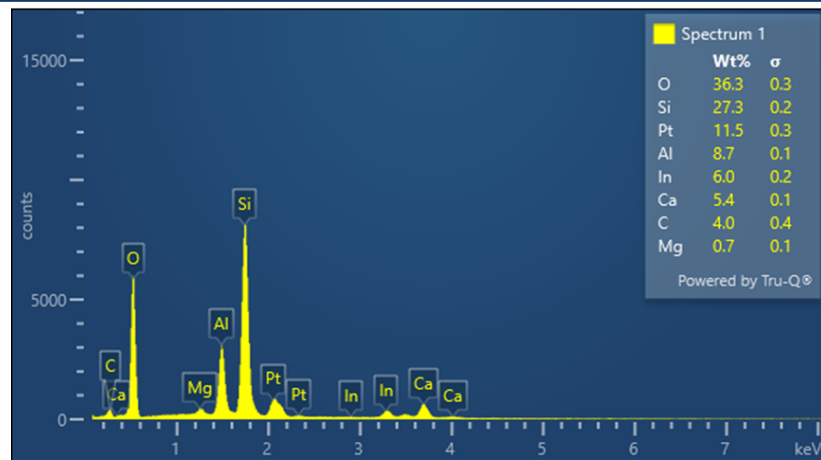
2026 (supplier A)

Appendix-2

3. Compared to the previous NG samples, the current samples show a lower oxygen content in the BS-ITO layer.
4. This indicates that the material condition is more controlled and does not exhibit the elevated oxygen level observed in the previous case.



2025 (supplier C)



oxygen content is approximately 36.3 wt%

2026 (supplier A)

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